Graph Ratio Tables

Lesson 3-3

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A + shape made by two number lines crossing at the center point (0, 0)

Coordinate plane

A grid with a line going across and a line going up to plot points.

(3, 6) means go right 3, up 6

Ordered pair

Two numbers (x, y) that tell where a point is on a grid.

(1,2), (2,4), (3,6) all line up

Linear pattern

Points that make a straight line on a grid.

A straight line from (0,0) through (2,6) and (4,12)

Proportional

Two amounts that grow together at the same rate.

The starting corner of the grid at (0, 0)

Origin

The point (0, 0) where the two grid lines cross.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can graph the values from a ratio table as points on the coordinate plane.

1️⃣ Notice

Chef Academy students are tracking how much chocolate sauce they need for different numbers of sundaes.

2️⃣ Key idea

I can graph the values from a ratio table as points on the coordinate plane.

3️⃣ Words I'll use

Coordinate plane

Ordered pair

Linear pattern

Proportional

Origin

🤔 Think About It

- What two quantities could we put on the x-axis and y-axis?
- What pattern do you see in the ratio table values?
- What do you think the graph will look like?

👀 Watch

Watch your teacher model one example. Jot what you see.

🤝 We try

Solve the next one together as a class.

✏️ You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The sundae recipe uses 2 ounces of chocolate sauce for every 1 sundae. Before you graph, what do you predict the points will look like, and which quantity goes on each axis?

Level 1 support

Start here: I predict the points will line up because sundaes and sauce grow together.

💡 **Need a hint?**

Sentence starters (optional)

I will put ____ on the x-axis and ____ on the y-axis.

Pondré ____ en el eje x y ____ en el eje y.

I predict the points will form a ____.

Predigo que los puntos formarán una ____.

Word bank: Coordinate plane | Ordered pair | Origin

Listen for: *A strong answer chooses sundaes (x) and ounces of sauce (y), and predicts a straight line because each sundae always needs 2 more ounces.*

Level 2

Predict whether the line will pass through the origin (0, 0). Justify your prediction using what 0 sundaes would mean.

- The line will pass through the origin because ____ sundaes means ____ ounces.
- I can justify this because the ratio stays ____ even at zero.

EXPLORE

After plotting (1,2), (2,4), (3,6)... what pattern do you notice, and how does it show the chocolate-sauce ratio?

Level 1 support

Start here: The points form a straight line, and each sundae adds 2 ounces of sauce.

💡 **Need a hint?**

Sentence starters (optional)

The points form a ___ through the ___.

Los puntos forman una ___ que pasa por el ___.

As the number of sundaes goes up by 1, the ounces go up by ___.

Cuando el número de copas sube en 1, las onzas suben en ___.

Word bank: Coordinate plane | Ordered pair | Linear pattern | Proportional | Origin

Listen for: *Listen for 'straight line through the origin' and the connection that the constant 2-ounce step shows a proportional relationship.*

Level 2

Critique this claim: 'Any points that go up to the right form a proportional graph.' What would the points (1,2),(2,4),(3,7) show instead, and why?

- The claim is wrong because a proportional graph must also ___.
- The points (1,2),(2,4),(3,7) are NOT proportional because ___.

CONNECT

The food truck plots tacos vs. salsa and gets a straight line. What does that graph tell the owner about preparing salsa for ANY number of tacos?

Level 1 support

Start here: The straight line lets the owner predict salsa for any taco amount.

💡 **Need a hint?**

Sentence starters (optional)

This is like our graphing work because ____ and ____ are related by ____.

Esto es como nuestro trabajo con gráficas porque ____ y ____ se relacionan por ____.

The graph lets the owner predict ____.

La gráfica permite al dueño predecir ____.

Word bank: Coordinate plane | Ordered pair | Proportional | Origin

Listen for: *A strong answer says the proportional line lets the owner read off (or extend to) salsa amounts for taco counts not in the table.*

Level 2

Generalize: if the food-truck line were steeper than another truck's line, what would that tell you about how much salsa each truck uses per taco?

- A steeper line means ____ salsa per taco because ____.
- I can compare the two trucks by looking at ____.

Guided Examples

Example 1

A ratio table shows (2, 6), (4, 12), and (6, 18). Which ordered pair comes next if the pattern continues?

Solution: The pattern adds 2 to x and 6 to y each time. After (6, 18): $x = 6+2 = 8$, $y = 18+6 = 24$. The next point is (8, 24).

Answer: A. (8, 24)

Example 2

When you graph equivalent ratios, the points will always form what shape?

Solution: Equivalent ratios are proportional, so their graph is always a straight line that passes through the origin (0, 0).

Answer: A. A straight line through the origin

Example 3

A ratio table shows (2, 8), (3, 12), and (5, 20). What is the y-value when x = 7?

Solution: The ratio is $y/x = 8/2 = 4$. For every x, $y = 4x$. When $x = 7$: $y = 4 \times 7 = 28$.

Answer: A. 28

Write About the Math

The Writing Revolution

I can explain my graph using the words coordinate plane, ordered pair, origin, and proportional.

1. Kernel Sentence

subject + verb

Model: Coordinate plane is a grid with a line going across and a line going up to plot points.
Plano cartesiano es una cuadrícula con una línea horizontal y una vertical para marcar puntos.

Write a kernel sentence about coordinate plane. Use a subject and a verb.
Escribe una oración base sobre plano cartesiano. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Coordinate plane matters in math
Plano cartesiano importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Coordinate plane matters in math because ____.
Plano cartesiano importa en matemáticas porque ____.

____

but
pero

Coordinate plane matters in math, but ____.
Plano cartesiano importa en matemáticas, pero ____.

____

so
entonces

Coordinate plane matters in math, so ____.
Plano cartesiano importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about coordinate plane.
Di un hecho verdadero sobre coordinate plane.

Coordinate plane ____.

Question
Pregunta

Ask a question about coordinate plane.
Haz una pregunta sobre coordinate plane.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about coordinate plane.
Muestra entusiasmo sobre coordinate plane.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with coordinate plane.
Dile a un compañero qué hacer con coordinate plane.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

I will put ____ on the x-axis and ____ on the y-axis.
Pondré ____ en el eje x y ____ en el eje y.

I predict the points will form a ____.
Predigo que los puntos formarán una ____.

The points form a ____ through the ____.
Los puntos forman una ____ que pasa por el ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Two students graph ratio tables. Student A plots (1,2), (2,4), (3,6). Student B plots (1,3), (2,6), (3,9). Whose line is steeper?

A. Student B

B. Student A

C. Same steepness

D. Cannot tell

Show your work:

2. A student plots the points (1, 3), (2, 6), (3, 9), and (4, 15) from a ratio table. She says they form a straight line because they go up. Is she correct? Explain how to check.

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Find Jada's Mistake — find the error, then write the correct reasoning.

Show your work:

Reflect — Exit Ticket

A ratio table shows cups of rice to cups of water: (1, 2), (2, 4), (3, 6). If you plot these points, the line passes through which point?

A. (5, 10)

B. (4, 6)

C. (5, 8)

D. (6, 10)

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Student B *— Student A's ratio is 1:2 (y goes up 2 for each 1 in x). Student B's ratio is 1:3 (y goes up 3 for each 1 in x). A higher rate of change means a steeper line, so Student B's is steeper.*
2. **Try It 2:** She is not correct. While the first three points follow the pattern y = 3x (3, 6, 9), the fourth point should be (4, 12), not (4, 15). If you plot all four points, (4, 15) would not land on the line through the others. The ratio 4:15 does not simplify to 1:3 like the others.
3. **Exit Ticket:** A. (5, 10) *— The ratio is 1:2, so for 5 cups of rice you need 5 × 2 = 10 cups of water. The point (5, 10) continues the pattern.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Coordinate plane is a grid with a line going across and a line going up to plot points.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).